Beyond Accuracy: An Empirical Study of Perception Stability in Multimodal Large Language Models

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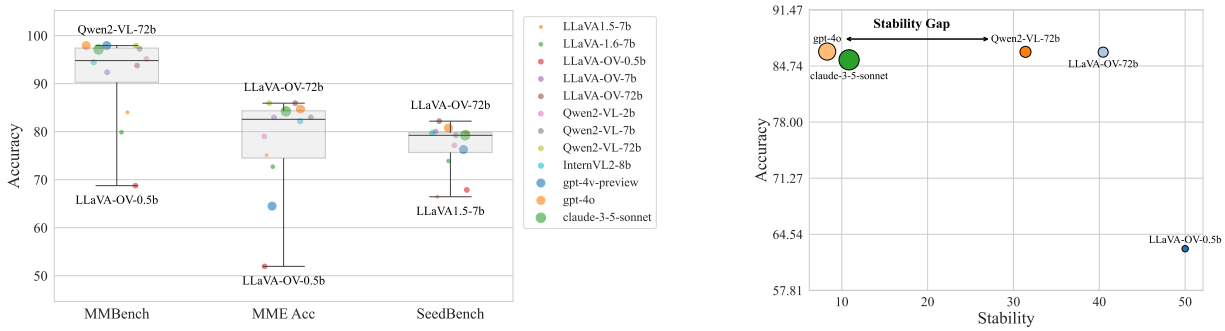


Figure 1. **Motivation and overview of our study.** Left: Best/worst models swap across MME [8], MMBench [36], and SeedBench [14] on entity capability, revealing instability that accuracy alone can mask. Right: We study perception stability in MLLMs, beyond accuracy, and reveal a substantial stability gap between open- and closed-source models. Sizes of closed-source models are approximate.

Abstract

Recent multimodal large language models (MLLMs) are predominantly evaluated on aggregate accuracy across perception benchmarks, with far less attention to how ***stable*** *their performance is across tasks, domains, and evaluation factors. This accuracy-centric focus yields a limited and sometimes misleading picture of multimodal perception capability. In this paper, we take a step toward systematically studying perception stability and analyzing how it scales across pre-training, post-training, and inference. We begin by decomposing 11 perception benchmarks into 6 core abilities and evaluating 18 MLLMs with two stability metrics that measure ranking volatility and performance dispersion across heterogeneous subtasks. Our findings show that perception stability is a fundamental yet long-overlooked dimension of MLLM behavior. Specifically: (i) although recent open-source MLLMs can match or surpass strong closed-source models in accuracy, they exhibit a substantial gap in perception stability; (ii) training dynamics reveal an ability conflict phenomenon, where gains in one or more abilities coincide with significant declines in another ability's accuracy and stability; and (iii) stability is shaped by different factors at different stages: during pre-training and post-training it is primarily driven by the data mixing ratio and LLM size, whereas at inference time it is strongly influenced by the model's reasoning style. Finally, targeted fine-tuning and model merging partially mitigate ability conflicts, but fully resolving multiple concurrent conflicts remains an open problem. Our code is available at* https://github.com/Chenfeng1271/AbilityLens.

1. Introduction

Recent multimodal large language models (MLLMs) [4, 15, 33] are primarily evaluated by aggregate accuracy on perception benchmarks, with far less attention to *perception stability*—the fine-grained consistency of performance across tasks, domains, and evaluation conditions. As shown in Figure 1, the best- and worst-performing models differ across MMBench [36], MME [8], and SeedBench [14] on enriry recognition, underscoring that accuracy alone can mask instability. This accuracy-centric view yields an incomplete and sometimes misleading picture of model capability, especially for real-world applications where consistency is critical. We

therefore investigate the fine-grained perception stability of MLLMs and examine how it evolves across stages of model development.

Our study begins by decomposing a broad-coverage suite of 11 image perception benchmarks [6, 8, 12–14, 20, 23, 32, 36, 40]—spanning diverse domains, instruction formats, question types, and metrics—into six core perception abilities (counting, OCR, attribute recognition, entity extraction, grounding, and structured-data understanding). We define perception stability as the consistency of a model's relative standing and absolute performance across heterogeneous subtasks (domains, formats, metrics), quantified by a rank-entropy score (RES) and a dispersion score (DVS), respectively. Building on this setup, we observe that although recent open-source MLLMs [4, 33] often match or surpass strong closed-source counterparts [9] in accuracy, they still exhibit a substantial gap in perception stability. To explain this discrepancy, we analyze how stability scales across training and inference and report three findings: (i) training dynamics reveal an ability conflict phenomenon—gains in one or more abilities coincide with significant declines in another ability's accuracy and stability; (ii) during training, stability is driven chiefly by data-mixture ratios and LLM size, with larger models and more balanced perception-focused mixtures tending to reduce conflicts; and (iii) at inference, a model's reasoning style (e.g., Chain-of-Thought [35], self-reflection [28], self-consistency [34]) strongly modulates stability, significantly improving performance consistency among multiple subtasks. Finally, we explore simple strategies to mitigate ability conflicts and improve performance: targeted fine-tuning and lightweight model merging yield partial gains, but fully eliminating multiple concurrent conflicts remains an open challenge.

Our contributions are as follows: (1) A principled formulation of *fine-grained perception stability* with two complementary metrics—RES and DVS—capturing ranking volatility and performance dispersion across heterogeneous subtasks. (2) A large-scale evaluation over 11 image-perception benchmarks, 6 abilities, and 18 MLLMs, revealing a persistent *stability gap* between open-source and strong closed-source models, and documenting an *ability conflict* during training. (3) A stage-wise analysis showing that stability is driven by data mixtures and LLM size during pre-/post-training, and by reasoning style at inference; simple fine-tuning and model merging partially mitigate conflicts but do not eliminate them.

2. Related Work

Perception-based MLLM benchmarks. In parallel with architectural advancements [4, 15], considerable effort has been dedicated to improving benchmarks for MLLMs, which play a crucial role in steering the development of the next generation of models [37]. As a fundamental capability, perception is essential to the performance of MLLMs. To evaluate this skill, benchmarks such as MME [8], MMStar [6], MMBench [36], and SeedBench [14] have been introduced, aiming to capture data from diverse domains and cover a wide range of question types. However, these differences inevitably introduce evaluation variance, motivating our study of perception stability.

Ability-perspective MLLM benchmarks. To enable a systematic evaluation of MLLM advancements, recent benchmarks [1, 25] aim to expose granular weaknesses within specific capabilities by regrouping benchmarks [16, 24, 37] according to model abilities. For instance, Unibench [1] categorizes 53 vision-language benchmarks by different capabilities, though most focus on traditional recognition tasks and are limited in evaluating state-of-the-art MLLMs. EUREKA [5], on the other hand, concentrates on non-saturated samples within each ability, offering a more targeted view of areas most ripe for improvement. However, these ability-centric efforts remain largely accuracy-oriented; we complement them with an ability-aligned evaluation that baseline-corrects heterogeneous submetrics and explicitly quantifies stability via rank-entropy (RES) and dispersion (DVS) across 18 MLLMs, enabling stage-wise analyses and revealing an open–closed-source stability gap.

Multimodal large language models. Building upon the success of LLMs such as GPT [29], LLaMA [31], and Qwen [3], and vision foundation models including DINOv2 [27], CLIP [30], and SigLIP [38], MLLMs [15, 18, 21, 22] have emerged as powerful tools for multimodal understanding and reasoning. However, training a versatile MLLM presents significant challenges [11]. Current state-of-the-art approaches [7, 17] primarily rely on more powerful LLMs and vision encoders coupled with extensive training data, requiring thousands of GPU hours for each training run. For example, Qwen2.5-VL [33] adopts a stronger LLM backbone and dynamic-resolution vision encoders, together with large-scale, multi-stage training (image–text pretraining, multi-task post-training, and RL-based post-training) on curated datasets. While these resource-intensive solutions have yielded improvements, we still lack a clear understanding of how these models evolve during training and how different abilities interact with each other. This opacity in model behavior impedes us to develop more effective strategies for advancing MLLM capabilities.

3. Perception Stability Evaluation

3.1. Dataset Decomposition

As illustrated in Table 1, we focus on 6 core perception abilities following prior works [8, 36]: counting, OCR, attribute recognition, entity extraction, grounding, and structural data understanding. We then select 11 relevant perception benchmarks to construct a uniform one for each ability. The in-

Table 1. **Decomposing 11 perception benchmarks into 6 perception abilities.** We reduce the sample size in large datasets since they have more redundant samples, where → indicates the number of samples before and after reduction.

Ability	Source	Subcategory	Question Type	Metric	Sample	Total Sample
Counting	MME	counting	T/F	mme_score	60	1759
	SeedBench	instances counting	MCQ	seed_img	2447→1223	
	MMStar	object counting	MCQ	fine_grained_perception	92	
	Muirbench	counting	MCQ	muirbench_score	234	
	Mirb	counting	VQA	mirb_score	150	
OCR	MME	ocr	T/F	mme_score	40	1844
	MMBench	ocr	MCQ	gpt_eval_score	304	
	OCRBench	ocr	VQA	accuracy	1000	
	Synthdog	ocr	VQA	tree_edit_distance	500	
Grounding	MME	position	T/F	mme_score	60	1493
	SeedBench	position, spatial relation	MCQ	seed_img	1636 → 817	
	MMStar	localization	MCQ	fine_grained_perception	40	
	MMBench	position, spatial relation	MCQ	gpt_eval_score	492	
	Muirbench	localization	MCQ	muirbench_score	84	
Entity	MME	landmark, celebrity	T/F	mme_score	740	2227
	SeedBench	identity reasoning	MCQ	seed_img	1831 → 915	
	MMBench	landmark	MCQ	gpt_eval_score	572	
Attribute	MMStar	attribute reasoning	MCQ	fine_grained_perception	188	3453
	Muirbench	difference spotting	MCQ	muirbench_score	536	
	MMBench	attribute recognition and comparison	MCQ	gpt_eval_score	405	
	SeedBench	instance attributes	MCQ	seed_img	4649 → 2324	
Structured Data	ChartQA	chart	VQA	relaxed_overall	2500 → 400	1332
	AI2D	graph	MCQ	exact_match	3088 → 300	
	ConBench	poster	VQA	ConScore_D	240	
	MMBench	code, map, diagram	MCQ	gpt_eval_score	282	
	MMStar	diagram	MCQ	fine_grained_perception	110	

Table 2. **Comparison between our decomposed dataset and existing perception-focused benchmarks.** Sample balance refers to the ratio between the highest and lowest sample counts across different capabilities.

Name	Samples	Sample Balance↓	Question Type			Performance	
			T/F	MCQ	VQA	Accuracy	Stability
MME [8]	2.37k	9.9	✓	✗	✗	✓	✗
SeedBench [14]	18k	50.8	✗	✓	✗	✓	✗
MMBench(en) [36]	11k	4.3	✗	✓	✗	✓	✗
ConBench [40]	3k	7.1	✗	✓	✗	✓	✗
UniBench [1]	-	-	✓	✓	✓	✓	✗
EUREKA [5]	83k	38	✓	✓	✓	✓	✗
Ours	12k	2.6	✓	✓	✓	✓	✓

cluded benchmarks encompass MME [8], MMBench [36], SeedBench [14], MuirBench [32], Mirb [41], MMStar [6], OCRBench [20], SynthDog [13], ChartQA [23], AI2D [12], and ConBench [40], covering diverse domains, question types and metrics. To reduce overlap and over-representation from large benchmarks, we follow LMMs-Eval [39] to remove redundant samples and perform deterministic subsampling on over-saturated categories, while preserving label balance and subcategory coverage. As shown in Table 2, this yields a much more balanced ability distribution, compared with prior ability-oriented suites that are heavily skewed toward a few recognition-style tasks.

3.2. Evaluation Metric

For each ability, we report two kinds of metrics—model *accuracy* and model *stability*—both computed from sub-metrics extracted from the source benchmarks. Because different question types have different chance levels (e.g., True/False has a 50% chance baseline by random selection, while multiple-choice depends on the number of options), we first align the chance baseline for every sub-metric so that ability scores are comparable across types. For each ability, let m_{ij} be the result of model i on sub-metric j, with UB_j the upper bound and BL_j the chance baseline[1]. The corrected score is

$$\tilde{m}_{ij} = \frac{m_{ij} - \mathrm{BL}_j}{\mathrm{UB}_j - \mathrm{BL}_j}. \tag{1}$$

Accuracy. Let n_j be the sample count of sub-metric j and $N = \sum_j n_j$. The accuracy of model i for this ability is the sample-count weighted average of the corrected sub-metrics:

$$\mathcal{A}_i = \sum_j \frac{n_j}{N} \tilde{m}_{ij}. \tag{2}$$

Stability. We consider two complementary measures: rank-entropy stability (RES) and dispersion-based variance stabil-

[1]For VQA, we set the baseline to 0.

Table 3. **Accuracy comparison.** Accuracy ↑ of 18 MLLMs where cells highlighted in green indicate the best performance, and red indicate the worst performance.

Model	Counting	OCR	Grounding	Entity	Attribute	Structured Data	Average
LLaVA1.5-7b	36.78	29.13	28.64	73.83	46.88	22.87	39.69
LLaVA1.6-7b	42.80	63.99	38.80	75.02	55.70	37.89	52.37
LLaVA-OV-0.5b	32.33	64.54	17.04	62.81	44.90	23.55	40.86
LLaVA-OV-7b	51.60	71.23	50.36	84.16	62.04	64.83	64.04
LLaVA-OV-SI-7b	49.44	78.17	45.24	85.55	62.22	61.40	63.67
LLaVA-Video-7b	37.82	53.08	32.00	68.56	60.63	33.13	47.54
LLaVA-OV-72b	56.75	81.35	59.23	86.40	69.08	73.15	70.99
Qwen2VL-2b	48.28	73.36	35.00	82.39	56.61	46.36	57.00
Qwen2VL-7b	50.95	79.29	47.12	85.10	62.66	67.56	65.45
Qwen2VL-72b	55.84	86.09	58.70	86.43	65.11	79.32	71.92
Qwen2.5VL-3b	50.27	77.49	40.01	82.56	60.70	56.10	61.19
Qwen2.5VL-7b	50.52	80.10	47.71	86.19	62.60	72.44	66.59
Qwen2.5VL-72b	56.53	85.70	59.62	86.82	67.11	79.20	72.49
InternVL2-8b	49.23	80.47	50.89	84.30	61.21	64.92	65.17
InternVL3-8b	51.12	80.02	51.45	84.55	63.46	70.12	66.79
gpt-4-vision-preview	39.00	71.95	47.67	77.92	58.56	67.12	60.37
gpt-4o-2024-08-06	50.55	82.57	58.84	86.47	67.67	75.82	70.32
claude-3-5-sonnet-2024102	51.64	79.80	58.63	85.47	61.20	77.06	68.97

ity (DVS). For RES, let $\mathcal{J}$ be the index set of sub-metrics for this ability and M the number of candidate models. For each sub-metric $j \in \mathcal{J}$, the M candidate models are ranked by the corrected score $\tilde{m}_{\cdot j}$ (larger is better). Let $r_{ij} \in \{1, \ldots, M\}$ denote the rank of model i on sub-metric j; ties are handled by competition ranking (e.g., $1, 2, 2, 4$). An empirical distribution p_{ik} of model i over rank positions k is defined by uniform averaging across sub-metrics:

$$p_{ik} = \frac{1}{|\mathcal{J}|} \sum_{j \in \mathcal{J}} \mathbf{1}[r_{ij} = k], \quad k = 1, \ldots, M. \tag{3}$$

where $\sum_{k=1}^{M} p_{ik} = 1$. Then, the Shannon entropy is

$$H_i = -\sum_{k=1}^{M} p_{ik} \log p_{ik}, \tag{4}$$

and the normalized rank-entropy stability $\mathcal{S}_i^{RES}$ is

$$\mathcal{S}_i^{RES} = \frac{H_i}{\log M} \in [0, 1], \tag{5}$$

where small values indicate more stable rankings across sub-metrics.

To assess the dispersion-based variance stability (DVS). of each ability, we first use z-score z_{ij} to measure the relative performance of model i with other model candidates in each sub-metric:

$$z_{ij} = \frac{m_{ij} - \mu_j}{\sigma_j}, \tag{6}$$

where μ_j and σ_j are the mean and deviation variance of the results $\tilde{m}_{\cdot j}$ of model candidates on this sub-metric, respectively. Then, we measure the standard deviation of the resulting z-scores as the DVS of model i in each ability, with higher values indicating greater instability:

$$\mathcal{S}_i^{DVS} = \mathrm{std}(\mathbf{z}_i). \tag{7}$$

Table 4. **Dispersion-based variance stability comparison.** DVS score ↓ of 18 MLLMs , with cells highlighted in red for the highest and green for the lowest scores.

Model	Counting	OCR	Grounding	Entity	Attribute	Structured Data	Average
LLaVA1.5-7b	72.54	128.40	42.32	110.40	85.30	46.73	80.95
LLaVA1.6-7b	48.00	72.73	31.47	57.29	28.37	43.95	46.96
LLaVA-OV-0.5b	69.20	54.25	60.61	50.00	82.58	73.21	74.00
LLaVA-OV-7b	65.09	62.05	49.28	26.81	72.56	18.62	59.64
LLaVA-OV-SI-7b	115.80	34.17	86.64	42.46	53.20	25.74	88.54
LLaVA-Video-7b	152.20	60.16	78.28	132.30	28.56	79.74	56.43
LLaVA-OV-72b	96.88	57.49	58.85	40.45	42.29	42.63	56.43
Qwen2VL-2b	47.50	89.28	56.53	20.97	64.92	46.80	54.33
Qwen2VL-7b	44.08	45.69	57.59	12.01	46.24	27.55	38.86
Qwen2VL-72b	88.18	24.78	51.52	31.40	49.34	27.82	44.56
Qwen2.5VL-3b	45.16	72.17	60.24	18.88	52.40	47.09	49.32
Qwen2.5VL-7b	45.10	46.70	56.91	14.09	47.28	22.17	38.71
Qwen2.5VL-72b	82.81	29.70	49.62	39.89	43.10	28.33	45.57
InternVL2-8b	44.16	17.34	47.93	9.74	54.60	60.48	38.54
InternVL3-8b	49.12	24.63	44.41	13.82	51.17	62.05	40.87
gpt-4-vision-preview	85.72	77.08	93.53	80.34	52.74	55.78	64.20
gpt-4o-2024-08-06	86.98	24.01	57.06	8.30	12.29	36.48	37.52
claude-3-5-sonnet-2024102	84.57	24.78	59.68	10.85	25.86	27.80	38.92

Table 5. **Rank-entropy stability comparison.** RES score ↓ of 18 MLLMs , with cells highlighted in red for the highest and green for the lowest scores.

Model	Counting	OCR	Grounding	Entity	Attribute	Structured Data	Average
LLaVA1.5-7b	41.34	72.53	24.36	62.99	47.40	26.29	45.82
LLaVA1.6-7b	29.31	41.21	17.75	34.39	18.03	26.40	27.85
LLaVA-OV-0.5b	39.29	63.09	35.57	28.02	47.84	42.79	42.77
LLaVA-OV-7b	37.82	34.86	28.60	15.44	42.34	11.90	28.49
LLaVA-OV-SI-7b	64.77	19.79	48.47	23.71	31.70	16.44	34.15
LLaVA-Video-7b	84.99	34.44	44.86	74.40	16.07	45.67	50.07
LLaVA-OV-72b	53.98	33.21	32.79	23.55	24.49	25.14	32.19
Qwen2VL-2b	28.79	51.78	33.93	12.40	36.39	27.19	31.75
Qwen2VL-7b	26.16	25.94	32.71	8.97	28.09	15.38	22.88
Qwen2VL-72b	50.96	15.20	30.46	18.44	29.14	15.76	26.66
Qwen2.5VL-3b	25.15	41.74	34.33	11.27	30.03	26.25	28.13
Qwen2.5VL-7b	27.75	26.07	33.06	9.34	26.44	14.08	22.79
Qwen2.5VL-72b	48.32	18.19	29.09	22.55	24.81	16.61	26.60
InternVL2-8b	25.12	10.11	27.14	7.64	31.24	35.01	22.71
InternVL3-8b	27.79	13.86	27.37	7.88	30.45	36.99	24.06
gpt-4-vision-preview	48.13	45.46	54.11	47.37	31.07	31.68	42.97
gpt-4o-2024-08-06	49.17	16.02	34.31	6.76	9.29	21.41	22.83
claude-3-5-sonnet-2024102	48.44	16.01	35.64	6.58	15.68	17.54	23.32

4. Main Results

We evaluate 18 MLLMs as model candidates with various model sizes and training data, including three strong commercial models: gpt-4-vision [26], gpt-4o [9], and claude-3-5-sonnet [2], as well as 15 open-source state-of-the-art models, including LLaVA 1.5 [17], LLaVA 1.6 [19], the LLaVA-OV series [15], the Qwen2VL series [33], Qwen2.5VL series [4] and the InternVL series [7]. The detailed version can be found in Table 3.

Observation 1. Open-source MLLMs often match or surpass closed-source models in accuracy but lag markedly in stability; their training pipelines remain accuracy-centric without considering stability objectives.

Overall stability. As shown in Table 3, 4 and 5, we find that closed-source models generally achieve better stability performance than open-source models, even attaining similar or worse accuracy levels. For example, Qwen2-VL-72b [33] has an average accuracy score of 71.92 and an average DVS of 44.56, while gpt-4o-2024-08-06 [9] has an accuracy of 70.32 and a DVS score of 37.52. This underscores the necessity of stability measurement to prevent model design

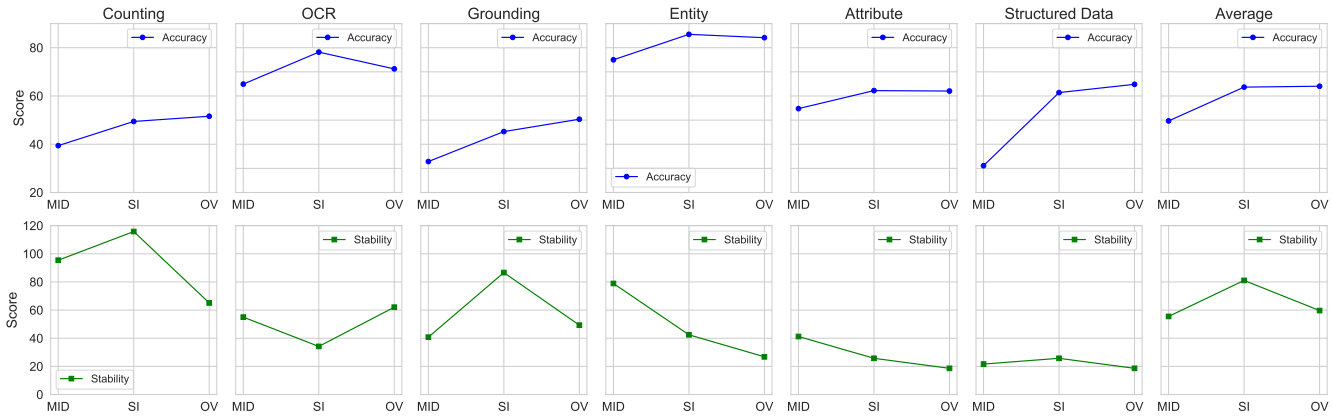


Figure 2. **Performance dynamics in the middle (MID), single-image (SI), and one-vision (OV) stages of LLaVA-OV-7b.** Accuracy curves demonstrate consistent improvement across stages, except for OCR tasks, whereas stability curves (DVS) indicate that stability is less prioritized in data preparation.

from overfitting single performance rankings. In RES, we find the performance is different from DVS, its mainly because model candidates are not enough. Moreover, we also examine the stability change in the multi-stage training of MLLMs. As shown in Figure 2, we analyze the performance dynamic between three stages of LLaVA-OV [15]. It reveals that current methods for developing training datasets are heavily focused on improving accuracy, while stability remains inconsistent and less prioritized.

5. Scaling Law of Stability

Because our pool of 18 MLLM candidates does not fully span the current performance landscape, we adopt dispersion-based variance stability (DVS) as the primary stability metric in this section, which is less sensitive to the number and composition of model candidates than rank-based measures.

5.1. Pre-training Scaling Law.

Observation 2. We observe an ability conflict phenomenon resulting poor performance: during training, gains in one or more abilities coincide with a significant decline in another ability's accuracy and stability performance.

Training dynamic. We exemplify the training scaling law by training LLaVA-OV-SI, which provides the full training procedure to develop an MLLM. As shown in the training dynamic of Figure 3, the OCR ability displays a marked performance degradation in both accuracy and stability at 10000 step, revealing an ability conflict that conventional methods might overlook.

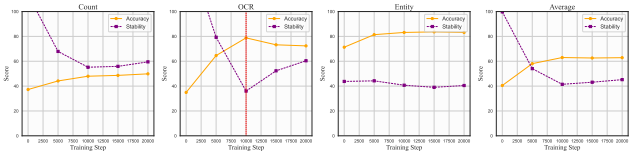


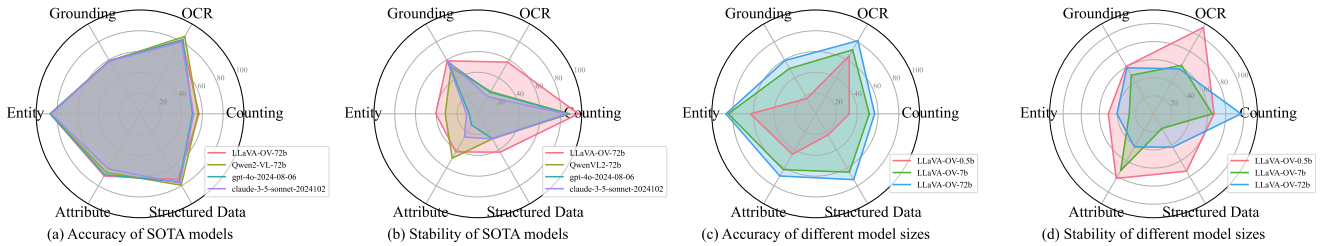
Figure 3. **Monitoring training dynamics of LLaVA-OV-SI.** Unlike traditional methods that focus solely on average scores, often obscuring underlying conflicts, we track optimization trends for each specific ability, revealing an ability conflict indicated by a significant performance degradation in accuracy and stability of OCR performance after 10,000 steps.

Observation 3. During pre-training, model stability is primarily driven by the data-mixture ratio and LLM size.

Reason behind ability conflict. To investigate the causes of ability conflict, we evaluate combinations of visual encoders, LLMs, and datasets (Figure 5, 6, 7). We find that conflicts are mainly attributable to model size and data mixing: different encoder–LLM pairings exhibit similar trajectories on the same dataset, whereas varying the mixture ratio or the LLM size markedly changes the magnitude, timing, and frequency of conflicts. In particular, smaller LLMs tend to incur more conflicts during training. These conflicts induce pronounced fluctuations in stability and often stall progress, making further improvement difficult despite continued optimization. Moreover, as shown in Fig. 4 (c)(d), increasing LLM size does not yield across-the-board gains at the ability level; unlike aggregate accuracy, the improvements are heterogeneous and concentrated in only a subset of abilities.

5.2. Post-training Scaling Law

We also try to improve stability by alleviating the ability conflict observed in Figure 3 in post-training manner, using



(a) Accuracy of SOTA models (b) Stability of SOTA models (c) Accuracy of different model sizes (d) Stability of different model sizes

Figure 4. **Accuracy and stability comparison** across AbilityLens's six perception dimensions, evaluating four state-of-the-art MLLMs alongside models of different LLM scales.

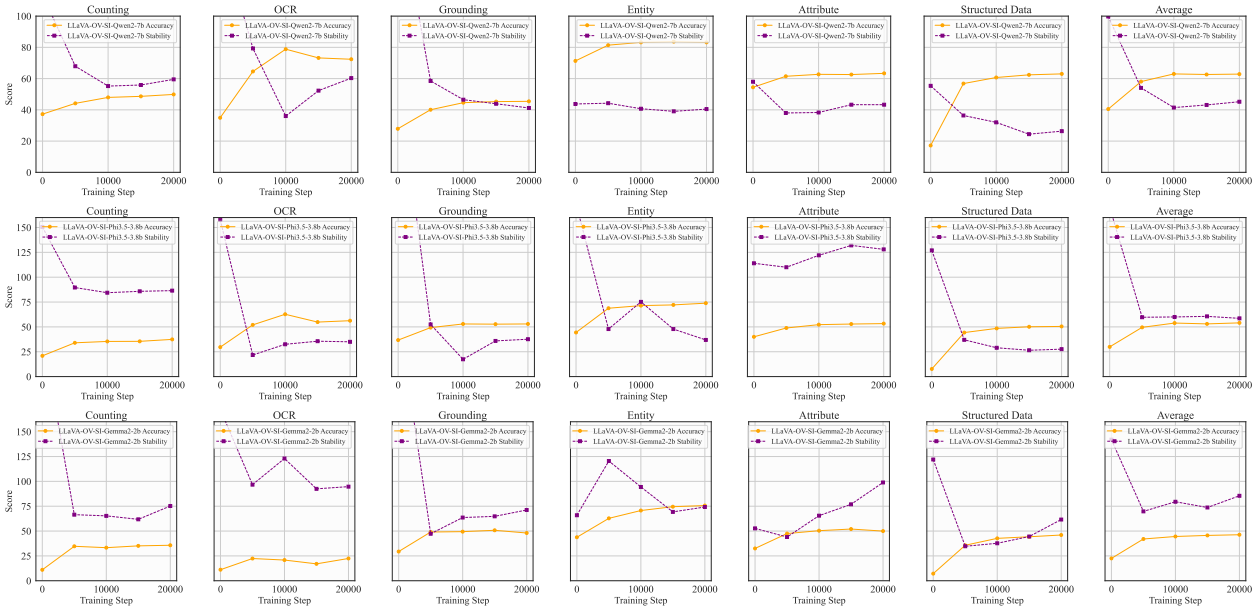


Figure 5. Monitoring the training dynamics of LLaVA-OV-SI using Qwen2-7b (first row), Phi3.5-3.8b (second row), and Gemma2-2b (third row) across six abilities. We find that model size significantly impacts the occurrence of ability conflict.

Table 6. Accuracy comparison ↑ with continuous fine-tuning and model merging methods in improving the weak OCR ability caused by ability conflict. Notably, the final step checkpoint is deemed as the baseline.

	GPU hours	Counting	OCR	Grounding	Entity	Attribute	Structured	Average
LLaVA-OV-SI-Qwen2-7b (10000 step)	-	47.96	78.82	44.58	83.18	62.73	60.70	62.99
LLaVA-OV-SI-Qwen2-7b (final step)	-	49.87	72.38	45.44	83.18	63.36	63.01	62.87
Continuous Fine Tuning	665.6	47.10	81.77	44.79	81.22	63.25	62.34	63.41
Task Arithmetic(addition) [10]	13.7	49.60	78.19	45.50	83.46	63.50	63.17	63.90

Table 7. Stability comparison ↓ with continuous fine-tuning and model merging methods in improving the weak OCR ability caused by the ability conflict.

	GPU hours	Counting	OCR	Grounding	Entity	Attribute	Structured	Average
LLaVA-OV-SI-Qwen2-7b (10000 step)	-	55.20	36.01	46.50	40.72	38.30	31.99	41.45
LLaVA-OV-SI-Qwen2-7b (final step)	-	59.55	60.30	41.22	40.48	43.31	26.35	45.20
Continuous Fine Tuning	665.6	62.19	45.76	43.39	45.60	48.11	24.51	44.93
Task Arithmetic(addition) [10]	13.7	59.19	37.72	46.19	41.96	44.27	26.08	42.57

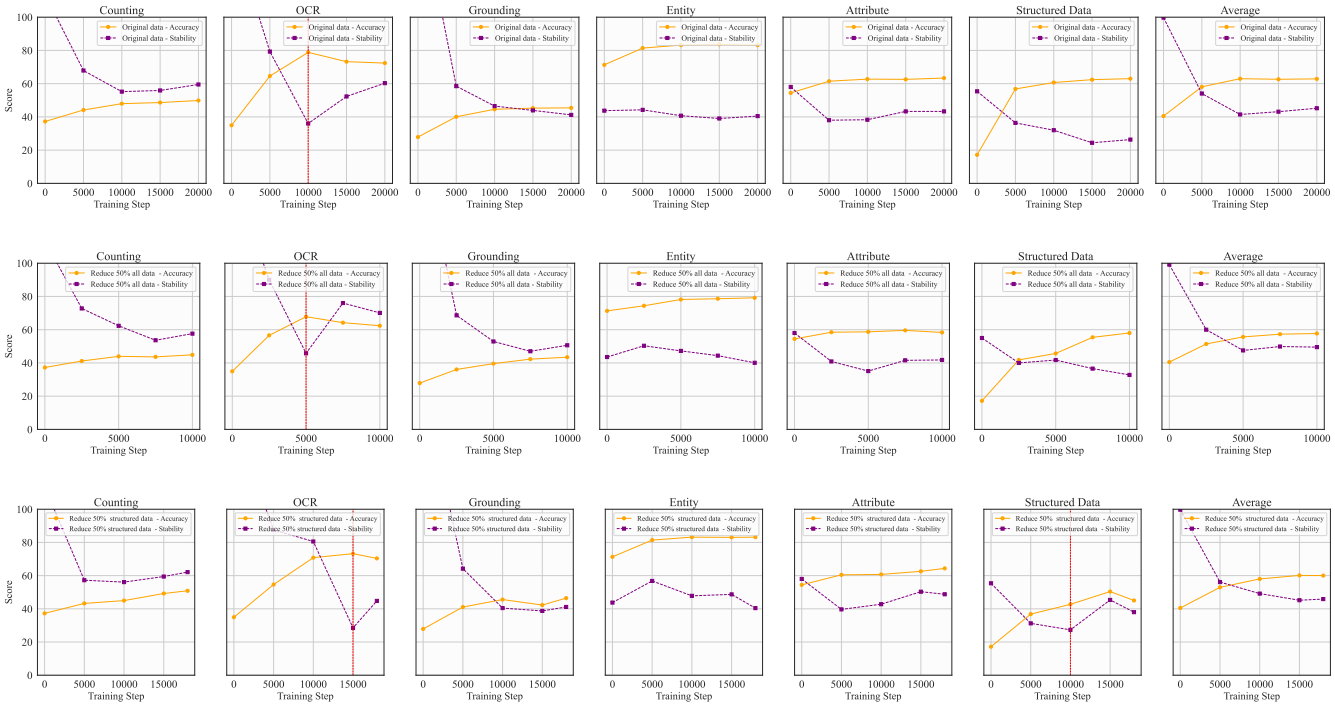


Figure 6. Monitoring training dynamics of LLaVA-OV-SI using original data (first row), 50% original data (second row), and original data reducing 50% structured data (third row) across six abilities.

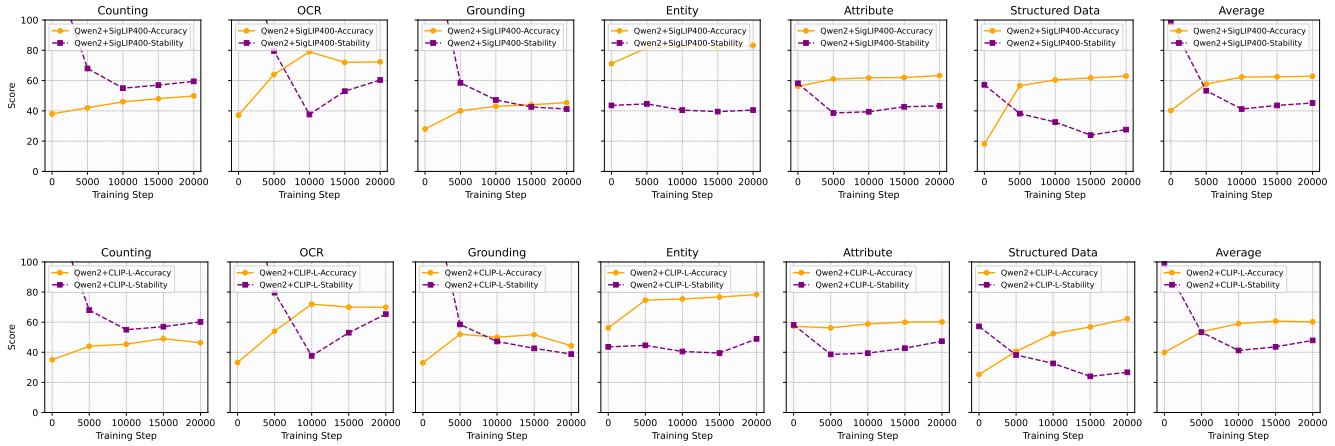


Figure 7. Monitoring the training dynamics of LLaVA-OV-SI-Qwen2-7b using CLIP-Large (first row) and SigLIP400m (second row) across six abilities.

continous fine tuning and model merging. For continuous fine-tuning of LLaVA-OV-SI-Qwen2-7b to enhance weaker capabilities, we retain the original training configuration and initialize the model from the final 20,000-step checkpoint. To mitigate catastrophic forgetting, we adjust the sampling ratio by setting general data to 10% and increasing OCR data to 20% for the single-image dataset of LLaVA-OV. For using model merging method Task Arithmetic [10] to alleviate ability conflict, we first collect a small calibration set with 3000 OCR samples from the single-image dataset of LLaVA-OV and tune the 1,0000-step checkpoint on it to find the task arithmetic. Then we add the arithmetic to the final checkpoint.

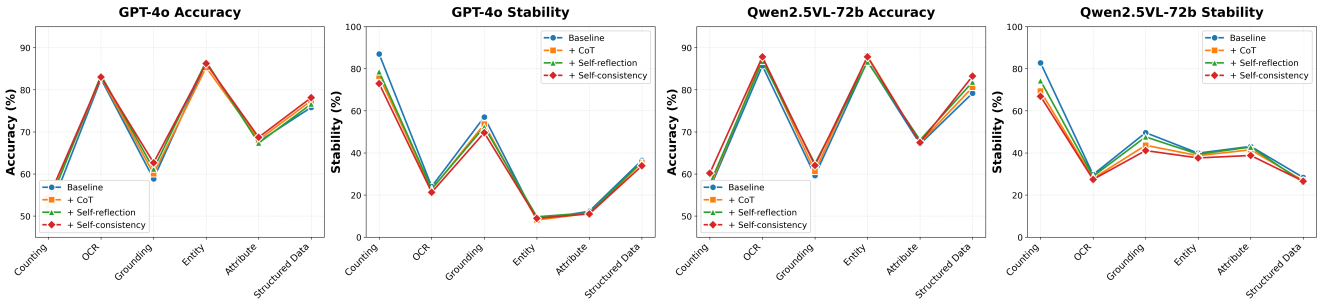


Figure 8. **Test-time scaling via reasoning on GPT-4o and Qwen2.5-VL-72B.** Accuracy (%) and stability (%) by ability under four inference settings: Baseline, +Chain-of-Thought (CoT) [35], +self-reflection [28], and +self-consistency [34]. Reasoning yields modest changes in mean accuracy but induces larger, ability-dependent shifts in stability.

Observation 4. Targeted fine-tuning and model merging can partially improve perception stability, but fully resolving multiple ability conflicts remains an open challenge.

Simple solutions to alleviate ability conflict. As shown in Table 7 and 6, when a conflict arises in a single ability during training, continuous fine-tuning or model merging can mitigate it. Task Arithmetic [10] generally offers a better accuracy–stability trade-off than continuous fine-tuning: it raises OCR to 78.19 (vs. 81.77 with fine-tuning), lifts the average accuracy to 63.90 (vs. 63.41), and reduces the average instability to 42.57 (vs. 44.93), while being less sensitive to data-mixture ratios. However, resolving conflicts that span multiple abilities remains open, as multi-model merging often fails to preserve each model's per-ability optimum without regressions.

5.3. Test-time Scaling Law

Observation 5. At inference, a model's reasoning style strongly modulates stability, sometimes improving consistency even when mean accuracy changes little.

We study reasoning as a mainstream form of test-time scaling, instantiated by three inference-time strategies: Chain-of-Thought (CoT) [35], self-reflection (critique–revise) [28], and self-consistency (sampling multiple reasoning paths with voting) [34]. We apply these to frozen models and keep prompts unified across abilities.

As shown in Figure 8, on both GPT-4o and Qwen2.5-VL-72B, adding test-time reasoning produces small, mostly positive accuracy shifts across abilities, whereas the stability changes are larger and ability-dependent. Self-consistency tends to be unchanged on OCR and structured-data abilities, while counting often shows better stability under reasoning. CoT delivers steady, modest improvements, and self-reflection helps curb brittle failures in several abilities, but effects vary by model. Overall, these results indicate a decoupling between accuracy and stability at inference time and highlight reasoning style as a strong knob for test-time scaling.

6. Conclusion

We studied perception stability of MLLMs beyond accuracy, decomposing 11 benchmarks into 6 abilities and evaluating 18 models with two complementary metrics (ranking and dispersion). Our results show: (1) open-source models, while often matching or surpassing closed-source in accuracy, still lag in stability; (2) stability scales differently from accuracy, with gains driven primarily by reducing ability conflicts; and (3) stability's drivers vary by stage—data mixing and LLM size during pre/post-training, and reasoning style at inference. Simple tactics (targeted fine-tuning, model merging) yield partial improvements, but resolving multiple concurrent ability conflicts remains open. We advocate reporting stability alongside accuracy and designing training/decoding strategies that explicitly optimize for stable perception. We hope this study will support researchers in drawing well-founded conclusions about effective strategies for advancing the fine-grained capability of MLLMs, avoiding overwhelming development testing.

Limitations and future work. Our study has several limitations. First, benchmark coverage remains restricted in domain breadth and metrics. Second, we focus primarily on perception abilities, leaving reasoning and instruction following largely unaddressed. In addition, our stability metrics (RES/DVS) capture ranking volatility and dispersion but not other reliability dimensions such as calibration or robustness under controlled perturbations. In future work, we will expand coverage to video/temporal, 3D, interactive, and multilingual settings with additional metrics (e.g., calibration, factor invariance, and tail-risk), enlarge the model pool with denser size checkpoints, and develop methods that explicitly optimize stability.

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